Invited Review

Updates on Testing, Treatment, and Prevention of Sexually Transmitted Infections in the United States, 2025

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Abstract: *There has been a substantial increase in the prevalence of sexually transmitted infections (STIs) and a widening gap in health care disparities over the past decade. New technologies, emerging public health research, and growing antimicrobial resistance have changed how practitioners counsel patients and diagnose and manage common STIs. Updating practitioners' understanding of best care practices for patients with STIs is crucial in averting preventable morbidity and mortality. This review covers important changes in the screening, diagnosis, treatment, and prevention of common STIs over the past 5 years.*

Keywords: antimicrobial resistance, bacterial vaginosis, chlamydia, extragenital testing, gonorrhea, *Mycoplasma genitalium*, point-of-care testing, sexually transmitted infection, STI vaccine, trichomoniasis

Introduction

Since 2009, rates of most bacterial sexually transmitted infections (STIs) have been steadily increasing in the US. However, recent trends show slight decreases in the rates of STIs overall, declining 1.8% from 2022 to 2023.[1] This decrease is most notable for gonorrhea, which saw a 7.2% decrease, with lower rates for all groups, but most significantly for women. In contrast, chlamydia rates remained stable and syphilis rates had a slight increase when including all stages and congenital syphilis.[1]

Unfortunately, disparities in STIs persist, with almost half of all reported STIs in 2023 diagnosed in adolescents and young adults under age 25 years. Men who have sex with men (MSM) are disproportionally impacted by all STIs, and HIV coinfection is common. One-third of gonorrhea, chlamydia, and primary and secondary syphilis cases were among non-Hispanic Black and African American individuals. Although these disparities reflect differences in sexual behaviors and networks, they also reflect differences in health care access and quality.[1]

With the substantial overall increases in prevalence and a widening gap in health care disparities, it is crucial to highlight recent updates on the diagnosis, management, and prevention of STIs. This need is further heightened by political changes in public health and health care funding, which highlights the importance of research and a forum to share advances. This review outlines major changes over the past 5 years for commonly diagnosed STIs in the US.

Updates on Screening and Diagnostics

Recent updates on the screening and diagnosis of STIs are included here, including strategies to increase testing, such as a focus on extragenital testing, screening for specific populations (eg, transgender and gender diverse), and opt-out testing. Additionally, updates in screening and diagnostics for *Mycoplasma genitalium* trichomonas, bacterial vaginosis (BV), and viral STIs are discussed. Given the scope of this review, syphilis and HIV are not included.

Extragenital Testing

Extragenital chlamydia and gonorrhea infections are common, although screening of the oropharynx and rectum is an emerging practice, especially for women.[2] A review of the literature found that among women,

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the median prevalence for rectal gonorrhea was 1.9%, pharyngeal gonorrhea was 2.1%, rectal chlamydia was 8.7%, and pharyngeal chlamydia was 1.7%. Among MSM, these numbers were much higher; median prevalence was 5.9% for rectal gonorrhea and 8.9% for rectal chlamydia.[2] Untreated extragenital infections are a potential source of ongoing transmission and may lead to an increased risk of HIV. Although screening strategies have evolved to recommend extragenital testing in MSM, routine screening for extragenital gonorrhea and chlamydia in all patients has shown substantial burden of disease in men who have sex with women (MSW) and women who have sex with men (WSM).[3] For women, extragenital screening increases the detection of gonorrhea or chlamydia by approximately 6% to 50% compared with urogenital screening alone. Standardization of extragenital screening for all individuals may markedly decrease STI transmission and morbidity because the majority of extragenital infections are asymptomatic and high-risk behaviors are not consistently reported.

Screening for Transgender and Gender-Diverse Individuals

Given the high prevalence of extragenital infections in transgender and gender-diverse individuals, the CDC recommends site-specific screening based on the patient's anatomy and sexual behaviors. It is important to identify terminology used by the patient to describe their anatomy and to mirror this language when discussing screening, diagnosis, and treatment. Annual gender-based screening for all sexually active females under age 25 years should include all individuals with a cervix, including transgender men and nonbinary persons. Transgender women with vaginoplasty sites should have routine STI screening for all exposed mucosal sites, and transgender men who have not had a vaginectomy should be screened for urogenital STIs from a cervical swab; a urine sample is not sufficient. Transgender individuals with HIV should be screened for all STIs at least annually.

Opt-Out Screening

Consideration of opt-out screening, particularly for adolescent and young adult females, was introduced in the 2021 CDC guidelines. Opt-out STI screening involves clinicians notifying patients that testing will be performed unless the patient declines. This is more effective in identifying infections than relying solely on patient disclosure of sexual behaviors, with one study demonstrating a decrease in chlamydia prevalence among young women by 55% utilizing an opt-out strategy.[4] An opt-out strategy has been demonstrated to be cost-saving among this population and may reduce the stigma of STI screening and diagnosis.[4] A study conducted in a county jail demonstrated that an opt-out approach to STI testing had a substantially higher prevalence and odds of testing positive for chlamydia than the standard opt-in strategy.[5] This was not shown for gonorrhea; however, this may be explained by overall lower rates of gonorrhea and a higher likelihood of patients with symptomatic infections who were excluded from the study.

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Point-of-Care Testing

There are several areas of active study for the diagnosis of STIs, most notably the development of point-of-care (POC) testing. Several POC tests are in development to reduce time to treatment and improve partner notification and expedited partner therapy (EPT). Several tests have been cleared by the US Food and Drug Administration (FDA) for gonorrhea, though only 1 is approved for extragenital infections and none detect pathogen resistance, which is crucial to minimize growing concerns of gonorrhea resistance.[6] Additionally, the FDA has cleared other POC tests for chlamydia and trichomoniasis. One at-home test for gonorrhea, chlamydia, and trichomoniasis (Visby Medical Sexual Health Test) has been FDA cleared with marketing authorization, which should help reduce stigma and increase accessibility of testing.

Updates on *M genitalium*, Trichomoniasis, and BV

M genitalium. *M genitalium* has been increasingly discussed as an underdiagnosed STI. As there are no population-based screening recommendations and it is not notifiable, there are few data on prevalence or natural history. Most cases of *M genitalium* appear to be asymptomatic. It is strongly recommended to avoid screening, especially with the increasingly utilized vaginitis panels; *M genitalium* is not a common cause of vaginitis, and positive test results can lead to overtreatment and contribute to growing macrolide resistance. Currently, the CDC recommends testing only in the case of persistent or recurrent urethritis, cervicitis, or pelvic inflammatory disease (PID) without another identified etiology.[7] Testing for *M genitalium* is currently by urine or mucosal nucleic acid amplification tests (NAATs), and molecular testing for resistance markers is not commercially available. No studies currently support extragenital testing.

Trichomoniasis. Trichomoniasis is not a reportable disease and thus there are no general screening recommendations. Currently, the CDC recommends only screening women with HIV given the increased prevalence in this population, first at entry to care and then annually. Historically, trichomoniasis was diagnosed by laboratory testing with wet-mount preparations to visualize the trichomonad, although NAATs are now preferred due to high sensitivity and specificity. Notably, these NAATs have been FDA cleared, but are not yet FDA approved.

BV. BV is a vaginal dysbiosis that is the most common cause of vaginal discharge worldwide.[8] Although not typically considered an STI, it has been understood that BV is associated with having numerous male sex partners, female sex partners, new or more than 1 sex partner, lack of condom use, and herpes simplex virus (HSV)-type 2 (HSV-2) seropositivity.[8] Individuals with BV are also at increased risk of STI acquisition. Diagnostics were historically laboratory based with a wet-mount and testing of the vaginal pH; however, NAATs have been increasingly used because of higher sensitivity and specificity than traditional methods.[8] Increasingly, POC tests have been utilized in the clinical setting to facilitate immediate treatment decisions, and self-swabs are acceptable and helpful for patient experience and buy-in.[9]

Updates on Viral STIs

HSV. Genital herpes is a chronic viral infection that can be caused by 2 different types of HSV: HSV-type 1 (HSV-1) and HSV-2. HSV-2 is responsible for most recurrent cases of genital herpes, although it has been increasingly recognized that an escalating number of genital herpes cases are caused by HSV-1, specifically among women and MSM.[1] Although HSV can be diagnosed clinically at the time of an outbreak, this can be challenging due to the self-limited and recurrent nature of genital herpes. If lesions are present, it is recommended to conduct virologic testing from the lesion by NAAT or culture to determine which type of HSV is causative. Overall, HSV-2 is much more likely to cause genital recurrence and subclinical shedding, and is associated with a 2-fold increase in HIV acquisition. Although previously identifying the subtype of genital HSV was not considered important, it is increasingly recognized that this testing can be informative for counseling patients and making decisions about prophylaxis and treatment. If there are no active lesions at the time of the exam, type-specific serologic testing can be used. However, these tests have low specificity, which can be improved with confirmatory testing by Western blot. Glycoprotein G testing is available to distinguish between HSV-1 and HSV-2 and has higher sensitivity and specificity. However, this test cannot determine timing or location of infection and can have high false-negative rates if done too soon following exposure. Glycoprotein G testing is not routinely performed currently in the US.

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Mpox. Mpox is an emerging STI with increasingly studied diagnostics and treatments. Although the initial outbreak in 2022 was characterized by systemic symptoms in addition to a painful rash, recent outbreaks have demonstrated anogenital skin lesions as the predominant feature.[10] There have been cases of severe mpox with 0.18% mortality in clade IIb, and these were primarily in immunocompromised patients, particularly those with HIV.[11] Currently, there are no indications for population-based screening, but the CDC recommends real-time PCR testing for potential mpox lesions (2 or 3 swabs in case additional testing is needed).[12]

Updates on Treatment

Over the past 5 years, there have been substantial changes in the treatment of STIs. Following are updates for the treatment of gonorrhea, chlamydia, *M genitalium*, trichomoniasis, BV, and viral STIs such as HSV and mpox.

Gonorrhea

Given substantial resistance to quinolones (35% of isolates), tetracyclines (28% of isolates), and azithromycin (5% of isolates), ceftriaxone is the mainstay of treatment for gonorrhea, with rare reports of resistance.[6] In 2010, the CDC recommended a single 250 mg intramuscular (IM) dose of ceftriaxone and a single 1 g oral dose of azithromycin for uncomplicated gonorrhea infection and to treat possible chlamydia coinfection. However, the CDC updated these guidelines in 2020 to recommend a 500 mg IM dose of ceftriaxone for uncomplicated gonococcal infections with a possible 7-day course of oral doxycycline 100 mg twice daily only if chlamydial infection could not be excluded.[13] There were several reasons for this change, the first being that of antimicrobial stewardship with several studies demonstrating the impact of recurrent antibiotics on the microbiome as well as resistance to azithromycin in other infections such as *Mycoplasma pneumoniae*, sexually transmitted enteric infections like shigellosis and campylobacteriosis, and nasopharyngeal pathogens such as *Streptococcus pneumoniae*.[14]

There are pharmacodynamic and pharmacokinetic reasons for increasing the recommended ceftriaxone dose. Models demonstrate that 500 mg is the current optimal dose to account for rising worldwide minimal inhibitory concentration (MIC) needed and body weight.[15] Ceftriaxone concentrations also tend to be more variable in the pharynx, resulting in a higher rate of ceftriaxone-based treatment failure for pharyngeal gonorrhea. As extragenital locations are tested less frequently, adjusting dosing to treat pharyngeal infections that may be missed will help decrease transmission.[16]

If ceftriaxone is unavailable for treatment of uncomplicated gonorrhea infections of the cervix, urethra, and rectum, the CDC recommends cefixime 800 mg orally once. Unfortunately, there are not reliable treatment alternatives for pharyngeal gonorrhea infections. Extragenital screening is especially important for gonorrhea as few regimens, other than ceftriaxone, will reliably cure infections of the pharynx and these infections are typically asymptomatic. For this reason, pharyngeal infections require a test of cure after 10 to 14 days.

If a patient has a cephalosporin allergy, the CDC recommends gentamicin 240 mg IM once plus azithromycin 2 g orally once. If chlamydia cannot be excluded and doxycycline cannot be used due to pregnancy, allergy, or concern for adherence, azithromycin 1 g orally once can be used. However, there remains concern of resistance with a substantial rise in necessary MIC over the past decade.[13]

Two novel antibiotics, zoliflodacin and gepotidacin, show promise for the treatment of gonorrhea given limited treatment options and rising concern for antimicrobial resistance. Zoliflodacin, a single oral dose, is especially promising for ease and adherence for uncomplicated urogenital gonorrhea, and FDA approval is anticipated in 2025. Gepotidacin is FDA approved for uncomplicated UTIs and has shown promise for the treatment of uncomplicated urogenital gonorrhea, though has not yet received FDA approval. A recent trial demonstrated that ertapenem 1000 mg IM is non-inferior to ceftriaxone for the treatment of gonorrhea.[6]

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Chlamydia

The CDC currently recommends doxycycline 100 mg twice daily for 7 days for the treatment of chlamydia. The previous recommendation, azithromycin 1 g orally once, is now a second-line treatment. Unlike for gonorrhea, these recommendations were not updated as a result of resistance concerns, but are related to treatment failure of rectal chlamydia with azithromycin. The CDC recommends a 21-day course of doxycycline for suspected or confirmed cases of lymphogranuloma venereum (LGV). Newer data suggest that a 7- to 14-day course of doxycycline or azithromycin 1 g weekly for 3 weeks may be adequate, but further research is needed.[17] A randomized trial for the treatment of rectal chlamydia demonstrated cure in 100% of individuals who were prescribed doxycycline, but only 74% of individuals who were prescribed azithromycin.[18,19] Treatment of rectal chlamydia is relevant for all populations based on a published review that noted that rectal chlamydia infection was detected among 33% to 83% of women who had urogenital chlamydia infection, independent of reported anorectal receptive sexual activity.[20] The prevalence of rectal infections regardless of reported sexual activity, and the differential antibiotic efficacy in various anatomic sites again highlights the importance of extragenital screening for all individuals.

Despite the noted superiority of doxycycline in treatment of chlamydia, compelling reasons remain for choosing azithromycin for the treatment of chlamydia. Azithromycin is a first-line treatment for those who are or may be pregnant and for those with an allergy or intolerance to doxycycline. Azithromycin is also a one-time dose that may overcome barriers to access, adherence, and confidentiality if given at POC in a clinical setting. Consideration of these factors necessitates conversations with patients about risks and benefits of each treatment option and prioritization of patient preference and goals.

M genitalium

The new recommendation for suspected or documented *M genitalium* urogenital infection is sequential treatment with doxycycline 100 mg twice daily for 7 days to reduce bacterial load followed by moxifloxacin 400 mg orally daily for 7 days. If local macrolide resistance is low or the strain is known to be macrolide sensitive, sequential treatment with first doxycycline followed by azithromycin 2.5 g over 4 days is recommended.[21] A resistance-guided approach highlights the importance of emerging POC testing for infections and for pathogen resistance.

> ***A recent randomized controlled trial demonstrated that treating male partners of individuals with BV with combined oral and topical antimicrobials resulted in substantially lower rates of recurrence than standard of care***

Trichomoniasis

A single 2 g dose of metronidazole was previously the standard treatment for trichomoniasis, and this remains the treatment for men and transgender women. However, the CDC now recommends a 7-day course of metronidazole (500 mg twice daily) for women and transgender men given the lower likelihood of treatment failure with this regimen.[22] EPT and retesting at 3 months had been recommended for women with HIV, although this is now recommended for all women given high rates of reinfection in the general population. In cases of treatment failure with low likelihood of reinfection, alternative regimens such as tinidazole (2 g daily) for 7 days can be considered.[23]

BV

Although BV is not considered an STI, recent understanding of the role of sexual behaviors and the benefits of partner treatment have led to changes in our conceptualization of vaginal dysbiosis. A recent randomized controlled trial demonstrated that treating male partners of individuals with BV with combined oral and topical antimicrobials resulted in substantially lower rates of recurrence than standard of care.[24] These novel findings are likely to substantially alter how BV is treated and characterized.

HSV

Treatment for genital herpes relies on suppressive or episodic treatment with nucleoside analogues, the first-line options being acyclovir, valacyclovir, and famciclovir. In pregnancy, daily prophylaxis starting at 36 weeks is recommended for all individuals with a history of any genital lesions. Additionally, all pregnant individuals should be questioned at the onset of labor about symptoms of genital herpes, including prodromal symptoms such as pain or burning before the appearance of a lesion.

Mpox

Although there are limited treatment data, it has been shown that most patients fully recover with supportive symptomatic care. Several antibody and antiviral treatments have been studied, although none have been demonstrated to be effective. Most recently, tecovirimat has been shown to be safe and well-tolerated, though ineffective in treating mpox.[25]

Updates on Prevention

In addition to recent updates on screening, diagnosis, and treatment, medications have been increasingly utilized in the prevention of STIs.

EPT

Currently, the CDC recommends EPT for all patients with chlamydia infection, recommends EPT with caution for individuals with gonorrhea infection where local resistance rates support it, and advocates for consideration of EPT for trichomoniasis when reinfection rates are high.[13,18,23] Expanded EPT has been demonstrated to reduce recurrent or persistent gonorrhea and chlamydia infections.[26] The CDC recommends the evaluation of all sex partners and empiric treatment of all partners in the past 60 days (or the most recent partner if there has been no contact for over 60 days). EPT is strongly discouraged for *M genitalium* given high levels of macrolide resistance and resistance-guided therapy. Historically, for MSM, the CDC recommended testing, and if needed, partner treatment given concerns of missed contact with the health care system and undiagnosed comorbid STIs such as HIV. Recent guidelines support the use of EPT for MSM through shared decision-making between the patient and the clinician to determine the safest and most appropriate approach.[27]

Doxycycline Postexposure Prophylaxis

Additionally, postexposure prophylaxis, specifically doxycycline (doxyPEP), is currently being studied as an effective strategy for reducing rates of gonorrhea, chlamydia, and syphilis infection. A recent landmark randomized clinical trial demonstrated a two-thirds reduction in rates of chlamydia and syphilis infection (although a less effective reduction in rates of gonorrhea infection) for MSM and transgender women who took doxycycline within 72 hours of condomless sex.[28] Data are not available for other demographic groups such as cisgender women. This is an ongoing area of active investigation, although it likely represents an important intervention for the prevention of bacterial STIs.

Vaccination

Several vaccines are in development to prevent STIs. A vaccine as primary prevention against gonorrhea infection could be vital in curbing gonorrhea resistance. Models predict that a vaccine that is 50% effective could eliminate 90% of gonorrhea infections if given to all 13-year-olds. Although a novel vaccine remains elusive, a recent study demonstrated that the commercially available MenB-4C vaccine was associated with reduced gonorrhea prevalence and may offer some cross-protection.[29]

A novel 2-dose modified vaccinia Ankara (MVA) vaccine exists and is safe and effective in preventing mpox infection. It is currently indicated for MSM at risk, including those with a new diagnosis of an STI, more than 1 sex partner, or sex at a commercial sex venue or event.[30] It is also indicated for a new or suspected exposure to someone with mpox infection. Pregnancy is not a contraindication to mpox vaccination. Data demonstrate that no booster dose is needed.[30] Lastly, an mRNA vaccine to prevent HSV recurrence is in phase I and II clinical trials.

Conclusion

With widening gaps in health care disparities and rising STI prevalence, it is increasingly important to highlight updates on the diagnosis and management of common STIs in the US. The fields of infectious diseases and public health are continuously evolving and updating our understanding of how best to care for patients with STIs is crucial in averting preventable morbidity and mortality.

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Financial relationships with ineligible companies within the past 24 months: Dr Neu reported serving on the Data and Safety Monitoring Board for Melinta Therapeutics. (Updated September 11, 2025) *Dr Katzman reported no relevant financial affiliations with ineligible companies.* (Updated September 11, 2025)

Reviewer/Planner 1 reported consulting or advisor fees from Generate Biomedicines and Gilead Sciences, Inc. (Updated November 26, 2025) *Planners/Reviewers 2 and 3 reported no relevant financial relationships with ineligible companies.* (Updated November 26, 2025)

All relevant financial relationships with ineligible companies have been mitigated.

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